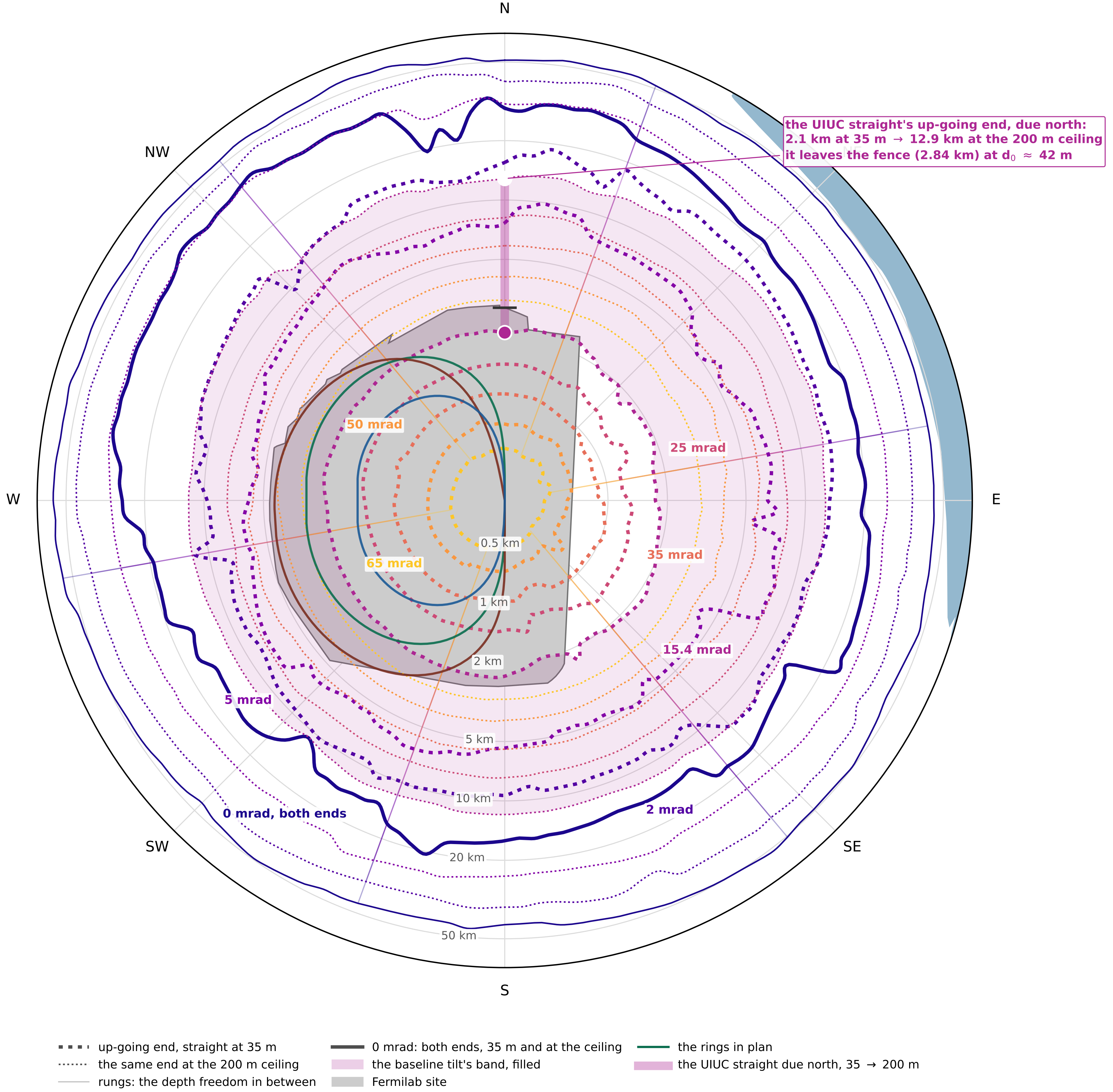
Near field, zoomed: where the up-going end surfaces for straights between 35 m and the 200 m ceiling
one colour per tilt; bold at 35 m, thin at the ceiling − $s_{near} \approx d_0/\theta$, so extra depth pushes every up-going exit outward



Deeper is farther: at the ceiling a level straight surfaces at 50 km both ways and a 65 mrad straight's up end at 3.1 km. The fence is 0.9−4.9 km out, so on-site emergence needs $d_0 \lesssim \theta\, s_{fence}$: about 42 m for the UIUC straight due north; at 50 mrad it is 45 m on the tightest bearing and the full 200 m everywhere the fence is past 4.0 km. Terrain: USGS NED 10 m inside 6 km, GEBCO beyond; 144 bearings, Catmull-Rom smoothed.